If this problem is ~~problem~~, $n = 100$
Then we need to compute all the subsets
of the set and find out the subset sum

$n = 100 \Rightarrow 2^{100}$ huge number.

This is a hard problem

Similarly prime factorization is also a hard problem when we consider classical computers.

mod 7

1 8 15 22 29

$(7x+1) \quad x \in Z \quad 7 \mid (29-8) = 21$

$= -6 \quad (7(-1)+1)$

modular arithmatic let 7 be the modulus.

So, remainders are. 0, 1, 2, 3, 4, 5, 6

$(7x_1+1) - (7x_2+1) \in Z \Rightarrow \{0, 7, 14, 21 \ldots\}$

$= 7(x_1 - x_2)$

$7x'$

$\{2, 9, 16, 23 \ldots\}$

8, 11

$8 + 11 = 19 \mod 7$

$= 5$

$8 \equiv 1 \mod 7$

$+ \quad 11 \equiv 4 \mod 7$

$\equiv 5 \mod 7$

$x \equiv 6 \mod 7$

$+ \quad y \equiv 5 \mod 7$

$\equiv 11 \mod 7$

$\equiv 4 \mod 7$

$x \equiv 1 \mod 7$

$= 30 \mod 7$

$= (28+2) \mod 7$

$= 2 \mod 7$

$= \ldots$

0, 1, 2, 3, 4, 5, 6

$x \equiv 5 \bmod 7$

$= (7a + 5)$ $a \in \mathbb{Z}$

$y \equiv 6 \bmod 7$

$= (7b + 6)$ $b \in \mathbb{Z}$

$x \cdot y = (7a + 5)(7b + 6)$

$= (7a \cdot 7b + 5 \cdot 7b + 6 \cdot 7a + 30) \bmod 7$

$= 30 \bmod 7$

$= 2 \bmod 7$

In Group

addition
multiplication
division

(G) Group If a non empty set. operation $*$

There are some groups among.

① $*$ binary operation $a, b \in G$ $\xrightarrow{\text{mod } 7}$

$a * b \in G$ $\quad x, y \in \xrightarrow{\text{mod } 7} Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$

$* =$ addition $\quad x + y \in Z_7$

② Associativity

$(a * b) * c = a * (b * c)$ $\quad (a+b)+c = a+(b+c)$

③ Identity $\quad e = 0$ $\quad 5 \bmod 7 + 0$

$a * e = e * a = a$ $\quad \{0, 7, 14, 21, 28, 35 \ldots\}$

④ Inverse $\quad 5 \bmod 7 + b = 0 \bmod 7$

$a * b$ $\quad = b * a = e$ or $b = -5 \bmod 7$ or $= 2 \bmod 7$

b is the inverse of a.

If my operation is $+$

$\Rightarrow$ my inverse is $-$

$G = (\mathbb{Z}, +)$ This is a group

$(x+y)+z = x+(y+z)$

$x+0 = 0+x = x$

$x+(-x) = (-x)+x = 0$

$x \cdot 1 = x$

$0 \cdot 1 = 0$

$x \times y$

$x, y \in \mathbb{Z}$

$x+y \in \mathbb{Z}$

$0 \in \mathbb{Z}$

$x \cdot y \in \mathbb{Z}$

$xy \cdot z = x(yz)$

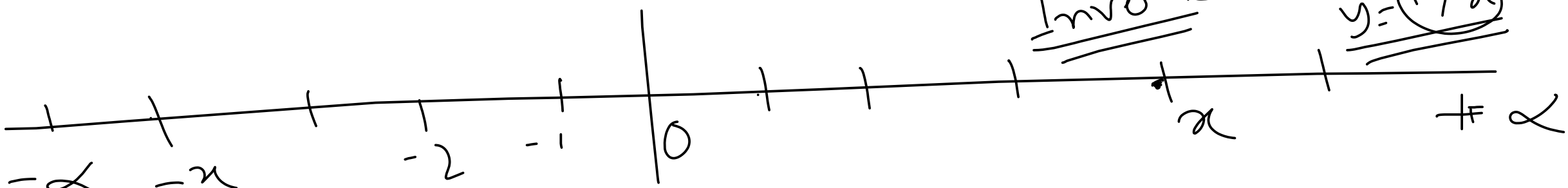
Identity

5, -5

Identity $= 1$

Inverse $x \cdot y = 1 \notin \mathbb{Z}$

$y = 1/x$



Q = Rational Numbers

$= p/q$ $\quad p, q \in \mathbb{Z}$ $\quad \underline{GCD(p,q) = 1}$

$\frac{1}{2} = \frac{1}{2}$ $\quad \pm\frac{2}{4} \pm\frac{3}{6}$ $\quad$ (24)

Q, $*$, we must remove 0

$I \cap Q$ a group under $\pm$, $*$

$\frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{p_1 q_2 + p_2 q_1}{q_1 q_2} \in Q$ $\quad p_1, p_2, q_1, q_2 \in \mathbb{Z}$

$= 1$ mult!

$\frac{0}{x} \quad \frac{0}{x} \quad x \in \mathbb{Z}\setminus\{0\}$ $\quad \frac{p_1}{q_1} + 0 = \frac{p_1}{q_1}$

$\frac{p_1}{q_1} \cdot y = 1$ $\quad y = \frac{q_1}{p_1} \in Q$ $\quad = -\frac{p_1}{q_1}$

$\therefore$ (0) multiplicative

Modular multiplication

$x = 2 \bmod 7$

$y = 3 \bmod 7$

$xy = 6 \bmod 7$

$Z_7 = \{0, 1, \ldots, 6\}$

$Z_m = \{0, 1, \ldots, m-1\}$

Identity $= 1 \bmod 7$

$x \cdot 1 = x$

$xy = 1 \bmod 7$

$y = x^{-1}$

$2 \cdot 2 = 4$

$2 \cdot 3 = 6$

$2 \cdot 4 = 8 = 1 \bmod 7$

$2 \cdot 5 = 3$

$2 \cdot 6 = 5$

$2 = 4^{-1}$

$4 = 2^{-1}$

$3 = 5^{-1}$

$5 = 3^{-1}$

$1 \cdot 1 = 6^{-1}$

$3 \cdot 1 = 3$

$3 \cdot 2 = 6$

$3 \cdot 3 = 2$

$3 \cdot 4 = 12$

$3 \cdot 5 = 15 = 1$

$1 \cdot 1 = 36$

$6 \cdot 6 = 36$

$= 1$

$2 = 4^{-1}$

$4 = 2^{-1}$

$3 = 5^{-1}$

$5 = 3^{-1}$

$1 = 1^{-1}$

$6 = 6^{-1}$

$\frac{6}{6} = 1 \quad \frac{5}{3} = 1$

mod 7

mod 8

$\mathbb{Z}_8^* = \{0, 1, 2, 3, 4, 5, 6, 7\}$

(2) $1 = 2$

$2 \cdot 2 = 4$

$2 \cdot 3 = 6$

$2 \cdot 4 = 8$

$2 \cdot 5 = 10 = 2$

$2 \cdot 6 = 4$

$2 \cdot 7 = 14 - 6$

$1 \cdot 1 = 1$

$3 \cdot 1 = 3$

$3 \cdot 2 = 6$

$3 \cdot 3 = 9 \cdot \mod 8$

$= 1 \mod 8$

3, 8 are co prime

$3 = 3^{-1}$

(5) $= 5^{-1}$

2, 8 are not co prime

2 has no inverse

$6 \cdot 1 = 6$

$6 \cdot 2 = 12$

$6 \cdot 3 = 18$

$6 \cdot 4$

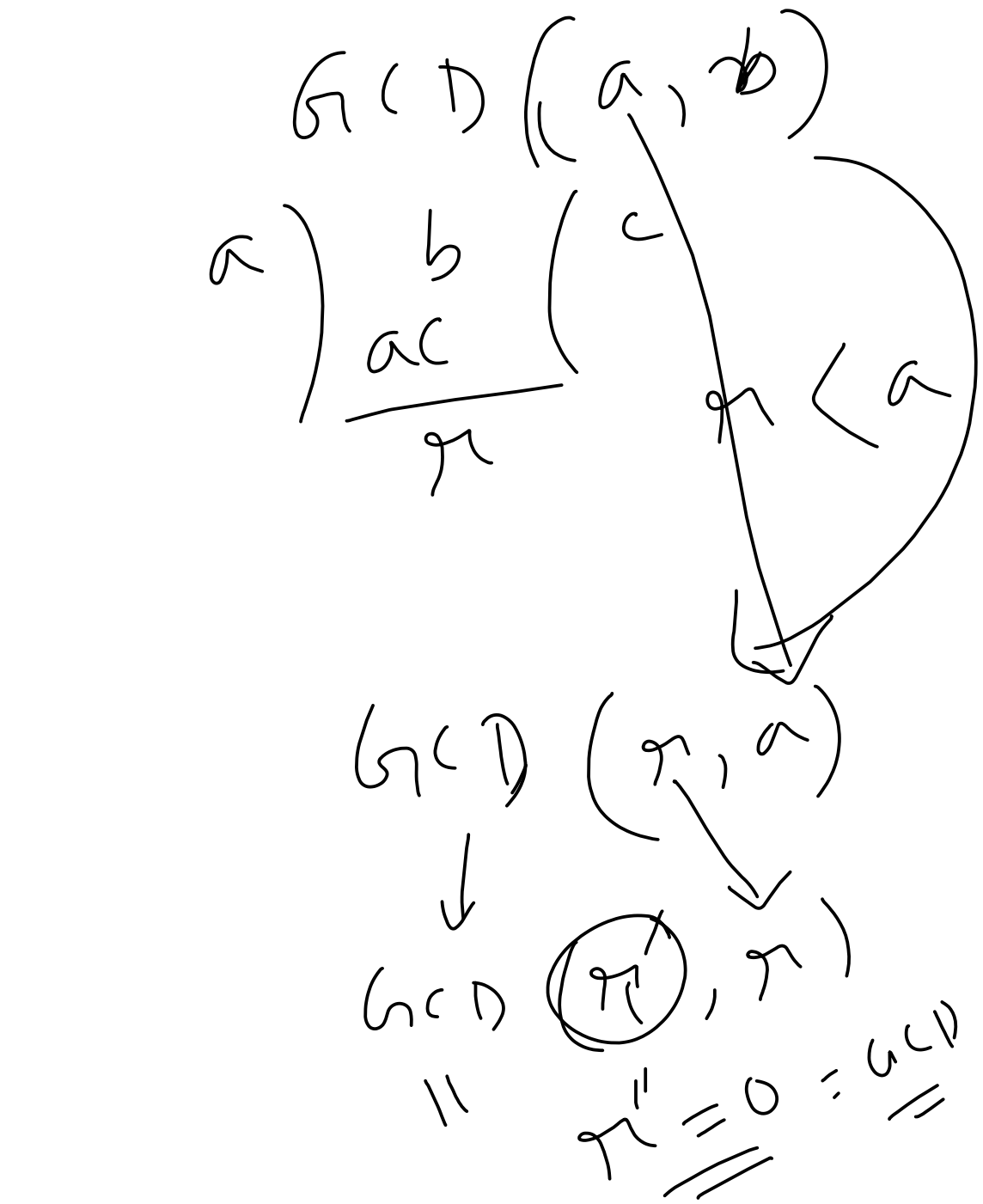
element

1

$7 \cdot 7 = 49$

when $\cdot = 1 \mod 8$

$\{1, 3, 5, 7\} = \mathbb{Z}_8^*$



b > a

8, 34

8) 34 (4
32
2) 8 (4 = 2
8
x

GCD = (8, 34)

2 is the maximum no that divides both 8 and 34

GCD (a, b) $\quad a > b$

$a = b \cdot q + r$

$r_{-1} = r_0 \cdot q_1 + r_1$

$r_0 = r_1 q_2 + r_2$

$\vdots$

$r_{n-2} = r_{n-1} q_n + r_n$

$r_{n-1} = r_n q_{n+1} + r_{n+2}$

r_{n-1} — r_n — GCD

r_{n+2} — 0

$r_{n-1} = r_n q_{n+1}$

$r_n \mid r_{n-1}$

$r_{n-2} = r_n + r_n q_{n+1} \cdot q_n$

$= r_n (1 + q_n q_{n+1})$

$r_n \mid r_{n-2}$

$r_n \mid r_{n-2}$

$r_0 = b$

$r_{-1} = a$

r_n divides both a and b

So, r_n is a common divisor of a, b

$r_n = \gcd(a, b)$

let $g = \gcd(a, b)$ $\quad g \mid a, g \mid b$

$r_{n-1} = r_0 q_1 + r_1 \Rightarrow g \mid r_1$

$r_{n-1} - r_0 q_1 = r_1$

$\Rightarrow g \mid r_2, g \mid r_n$

r_n divides both

g is the gcd

So, $\Rightarrow g = r_n$

$r_n = 2g$ some common divisor

$2 \mid a,$

$2 \mid b$

g_n is found by the eclidian Algo.
and this is the GCD of (a, b)

$$gcd(a,b) = ax + by \quad \forall x, y \in \mathbb{Z}$$

$a = 3$
$b = 7$

$$3x + 7y = 1 \qquad x = -2, \quad y = 1$$

$$\begin{array}{r|l} 3 & 7 \\ 6 & 2 \\ \hline 1 & 3 \\ 3 & \\ \hline x & \end{array}$$

$gcd(1)$

$3(-2) + 7 \cdot 1$
$= -6 + 7 = 1$

$$\gcd(6, 14)$$

$$= 2$$

$$2 = \frac{6x + 14y}{2(3x + 7y)}$$

$$= 2(3x + 7y)$$

$$= 2(-6 + 7) = 2$$

$$x = -2 \quad y = 1$$

$$6 \overline{)14} (2$$
$$12$$
$$\overline{2)6} (3$$
$$6$$
$$\overline{0}$$

$$x = -2$$

$$y = 1$$

Extended Euclid Algorithm

$8)13(1$
$\quad 8$
$5)8(1$
$\quad 5$
$3)5(1$
$\quad 3$
$2)3(1$
$\quad 2$
$1)2(2$
$\quad 2$
$\quad\ \ 0$

$1 = 3 - 2 \cdot 1$
$= 3 - (5 - 3 \cdot 1) \cdot 1$
$= 3 \cdot 2 - 5 \cdot 1$
$= (8 - 5) \cdot 2 - 5 \cdot 1$
$= 8 \cdot 2 - 5 \cdot 3$
$= 8 \cdot 2 - (13 - 8) \cdot 3$
$= 8 \cdot 5 - 13 \cdot 3$
$= 8(5) + 13(-3)$

$x = 5$
$y = -3$

$\gcd(a,b) = ax + by$
$x, y \in \mathbb{Z}$
= function of a, b